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Does Movement Change When Low Back Pain Changes? A Systematic Review

Low back pain (LBP), the most burdensome nonfatal health condition,^{16,17} is highly prevalent¹⁴ and expected to have an increasing impact.³ Nonspecific low back pain (NSLBP) describes back pain where no definitive pathoanatomical cause can be determined, which is approximately 90% of the time.^{28,37}

Patients, clinicians, and researchers frequently believe that the way a person moves may influence the likelihood of developing and recovering from NSLBP.^{26,34,36} Movement interventions²⁶ are predicated on effecting a change in movement to improve back pain. Slower movements with less multiplane range of motion (ROM), less variability, increased stiffness, and altered patterns of trunk muscle activity characterize people with

NSLBP.^{13,22,29,31,44,52,67} However, it is unclear whether changes in the way people with NSLBP move are related to changes in pain and activity limitation.

Concurrent changes in lumbopelvic movement and improvements in pain or activity limitation are infrequent.³² However, previous research has typically evaluated group-level change; therefore, it is unclear whether the people who changed their movement were also those



who had a change in pain or activity limitation (individual-level change). Conclusions based on group-level results can be different from those based on individual-level results.^{23,41,54}

A systematic review of individual-level data only found a relationship between changes in spinal flexion, lateral flexion, and rotation ROM and changes in pain or activity limitation in 1 in every 3 (6.5/19) of the associations investigated.⁶⁴

Current research does not support the deeply held belief of a relationship between changes in the way someone moves and a change in back pain, although movement parameters (ie, velocity and muscle activity) have not been comprehensively investigated.

Thus, (1) synthesizing data from studies that measure individual-level changes in both movement and pain or activity limitation over time, and (2) data that include diverse movement parameters (such as movements in additional planes, lumbopelvic patterns, velocity and muscle activity parameters) may help to clarify the relationship between changes in movement and changes in pain or activity limitation at the individual level.

Therefore, the aim of our review was to determine whether there was a relationship between changes in volitional spinal movement (lumbopelvic kinematics, lumbar flexion/extension angle, lumbar flexion during functional tasks, lumbar

• **OBJECTIVE:** To investigate the relationship between changes in volitional spinal movement (including muscle activity) and changes in pain or activity limitation at the individual level in people with nonspecific low back pain.

• **DESIGN:** Etiology systematic review.

• **LITERATURE SEARCH:** MEDLINE, Embase, CINAHL, and AMED were searched from inception to January 2020.

• **STUDY SELECTION CRITERIA:** The study included peer-reviewed articles that reported the relationship between changes in volitional spinal movement and changes in pain or activity limitation at the individual level in people with nonspecific low back pain.

• **DATA SYNTHESIS:** The data were descriptively synthesized to identify a relationship between change in movement and improved pain or activity limitation.

• **RESULTS:** We included 27 studies involving 2739 participants. There was low-quality evidence of a relationship between change in movement and change in pain or activity limitation at the individual level 31% of the time (20 of the 65 times investigated within the 27 studies). Increases in spinal range of motion, velocity, and flexion relaxation of the back extensors were consistently related to improved pain or activity limitation (93%, 18.5/20 relationships observed).

• **CONCLUSION:** A relationship between changes in movement and changes in pain or activity limitation was infrequently observed at the individual level; however, a paucity of high-quality evidence precludes a definitive understanding of this relationship. *J Orthop Sports Phys Ther* 2020;50(12):664-680. Epub 28 Oct 2020. doi:10.2519/jospt.2020.9635

• **KEY WORDS:** exercise therapy, low back pain, movement, physical therapy, systematic review

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contribution to flexion, velocity and timing of lumbar kinematics, flexion relaxation, lumbopelvic muscle activity) and changes in pain or activity limitation at the individual level in people with NSLBP.

METHODS

THE PRISMA STATEMENT⁵⁰ WAS FOLLOWED for reporting items in this prospectively registered systematic review (PROSPERO registration: CRD42017064436).

Identifying and Selecting Studies for Review

Information Sources The electronic databases from MEDLINE, Embase, CINAHL, and AMED were searched from inception to August 21, 2017. An updated literature search was performed on January 8, 2020. Reference lists of retrieved articles were screened.

Search Our comprehensive search strategy was adapted from a previous systematic review.³² We used medical subheadings and key words that encompassed the themes of changes in movement and changes in pain and activity limitation in people with NSLBP (see **APPENDIX A** for the search strategy used in the MEDLINE database). Our search strategy was developed in collaboration with a university faculty librarian. The strategy included 2 previous systematic reviews in the field^{32,64} that were also assessed.

Selection Criteria We included studies of the relationship between changes in volitional spinal movement and changes in either pain or activity limitation at the individual level in people with NSLBP. Low back pain was defined as pain between the 12th thoracic vertebra and gluteal folds. There was no restriction on the duration of LBP. Interventions had to be nonsurgical and nonpharmacological, but could be cognitive, behavioral, physical, or of a combined nature targeting the body and/or mind.

We excluded studies that included pregnant women or people who had a

specific pathological cause for their back pain (such as spinal malignancy, infection, fracture, cauda equina syndrome, metabolic or spinal inflammatory disorders, or nerve compression). Studies had to be written in English, and participants had to be older than 12 years of age.

Study Selection We used Covidence systematic review software (Veritas Health Innovation Ltd, Melbourne, Australia) to manage study selection. Following the removal of duplicates, eligibility assessment was performed on a standardized form independently by 2 reviewers (K.W. and J.T.), who were blinded to each other's assessments. Conflicting assessments were resolved by discussion or by a third reviewer (P.K.) when consensus could not be reached. Titles and abstracts, as well as full-text versions of the included studies, were screened using the same procedures.

Quality Assessment

Risk of Bias Within Individual Studies Our aim was to determine the prevalence of statistical associations between change in one patient characteristic and change in another. The sources of bias relevant to questions about these characteristics, treatments, and etiologies overlap but are not identical. Although all participants included in our systematic review received treatment, treatment exposure was not relevant to our research question. The included studies estimated the strength of a relationship between change in movement and change in pain or activity limitation. Due to heterogeneity of movement parameters, measurement tools, interventions, and analysis techniques in the included studies, we could not pool the data. Instead, we used vote counting to report the prevalence of a significant relationship across the included studies; therefore, some sources of bias were different.

The most important sources of bias in our systematic review were related to how well the population was described (generalizability of the findings), the validity and reliability of the method of measure-

ment of the variables, the missingness of data, the appropriateness of the statistical analysis, and assessor blinding. We adapted the Joanna Briggs Institute critical appraisal checklist for cohort studies⁵¹ to reflect these sources of bias (**APPENDIX B**). While no psychometric data are available specifically for this tool, the original checklist for cohort studies was extensively peer reviewed,⁵¹ and a similar tool has satisfactory validity.⁵³

We pilot tested and adapted the tool before 2 reviewers (K.W. and J.T.) independently assessed each study included in our systematic review. Conflicting assessments were resolved by discussion or by a third reviewer (P.K.) when consensus could not be reached. The independent ratings of the 2 reviewers had a percentage agreement across the 9 assessment items of 92.3% to 100.0% (median, 96.2%; interquartile range, 2.9%).

Studies that scored positively on all checklist items had a low risk of bias. Because the most critical risks of bias for our research question were related to the validity and reliability of the method of measurement of the 2 variables, the appropriateness of the statistical analysis, and assessor blinding, we categorized studies as having a medium risk of bias when they scored positively on all those items and negatively or were unclear on noncritical risks. A study was at high risk of bias when any of the critical-risk items were scored negatively or were unclear.

Quality of Evidence Across Studies We used the Grading of Recommendations Assessment, Development and Evaluation (GRADE) tool¹⁸ to rate the overall quality of evidence across the studies. Risk of bias, imprecision, inconsistency, indirectness, and publication bias were assessed relative to our research question about the presence of a relationship between change in movement and change in pain or activity limitation. Randomized controlled trials (RCTs) were analyzed as cohort studies, because treatment allocation was not relevant to our research question. The GRADE guidelines suggest starting at "medium quality" for research

using cohort designs. We used the other GRADE criteria to upgrade or downgrade the quality score (further detail on our application of the GRADE tool can be found in **APPENDIX C**).

Data Analysis

Data-Collection Process and Data Items

A data-extraction table was pilot tested on 4 studies and refined to improve efficiency in collecting key outcomes related to our research question. Subsequently, greater emphasis was placed on extracting details of the movement and pain and activity limitation measures, so that their validity and reliability could be better evaluated. For all included papers, 2 reviewers (K.W. and J.T.) independently extracted the following data: (1) study design, (2) participant characteristics, (3) sample size, (4) intervention (including type, dosage, duration, and frequency), (5) movement parameters, their change, and method of measurement, (6) clinical outcome (pain or activity limitation), its change, and method of measurement, and (7) statistical analysis method and results about the relationship between changes in movement and changes in outcome (if available). Conflicts were resolved by discussion and, when consensus could not be reached, a third reviewer (P.K.) was consulted. Three authors^{6,11,60} were contacted via e-mail for further information but did not respond.

Types of Movement Measures Volitional spinal movement measures encompassing either spinal kinematic or muscle activity parameters that were objectively quantified using an instrumented technique were included. Acceptable measures included (1) lumbopelvic kinematics (eg, lumbar flexion/extension angle, lumbar flexion ROM during functional tasks, lumbar contribution to flexion, velocity and timing of lumbar kinematics) and (2) muscle activation (eg, the flexion-relaxation phenomenon, lumbopelvic muscle activity during volitional spinal movement). The flexion-relaxation phenomenon refers to the electrical “silence” (relaxation) of the paraspinal

muscles when an individual reaches end-range flexion.⁷⁰ This period of relaxation is frequently absent in individuals with back pain, and a flexion-relaxation ratio (largest in people without LBP) has been developed to standardize reporting of the flexion-relaxation phenomenon.⁷⁰

Measures were defined as representing lumbar ROM if the calculation was derived from a lower thoracic or upper lumbar region relative to a pelvic segment; otherwise, they were classified as multisegmental measures. Parameters related to measures not including volitional spinal movement (such as gait, proprioception, muscle timing and cross-sectional area during static positions, or feedforward onset in response to alternate limb movement) were excluded.

Timing of Measurement Movements had to be measured at a minimum of 2 time points to obtain a measure of change. If multiple time points were reported (eg, postintervention and 3- and 12-month follow-ups), the time point closest to the end of the intervention period was extracted.

Primary Measure The primary measure determined the presence of a relationship between a change in movement and a change in pain or activity limitation at the individual level. In addition to the presence of a relationship, we also investigated the direction of the relationship, for example, whether an increase or decrease in movement ROM, velocity, or flexion relaxation was related to improved pain or activity limitation. We also combined the pain and activity limitation data into one outcome to provide a supplementary, broader summary of the relationship between changes in movement and changes in outcome.

Synthesis of Results

A descriptive synthesis was undertaken, with a sensitivity analysis planned based on the risk of bias of the individual studies.

We used a vote-counting system to summarize how frequently a relationship was observed. Each movement parameter was assigned 1 vote (value of 1.0) when a relationship was found between the pa-

rameter and pain or activity limitation. For studies that used multiple measures for a single movement parameter (eg, measuring flexion with both a modified-modified Schober test and through rachimetry with computer analysis¹⁰), the vote was divided by the total number of measures used.

People with high levels of pain and activity limitation may have a greater capacity to change their pain or activity limitation and, in turn, may be more likely to show a relationship with concomitant changes in movement. Therefore, as an additional analysis, we assessed whether (1) baseline levels of pain and activity limitation were comparable to those in published NSLBP trials, and (2) mean baseline levels of pain and activity limitation were comparable in studies that did and did not identify a relationship with flexion-related movement, the most common movement investigated. All pain and activity limitation values were transformed to a 0-to-100 scale (higher scores are worse). If a study reported multiple flexion parameters, that study was only counted once in each group (either it did or did not show a relationship) to avoid counting the same participants more than once (to preclude bias toward studies that measured multiple flexion parameters).

RESULTS

THE SEARCH IDENTIFIED 3920 ARTICLES, of which 1104 were duplicates; therefore, we assessed 2816 articles for eligibility. The reasons for exclusion are presented in the PRISMA flow diagram (**FIGURE 1**).

Characteristics of Included Studies

We included 27 studies^{2,6,9-12,19,23,24,35,38-42,45-47,49,54,59-62,66,69,73} (n = 2739 participants). Seven studies^{24,35,38-40,49,60} used data from an RCT. The details of individual studies are presented in **APPENDIX D** (available at www.jospt.org), while **TABLE 1** presents summaries related to sample size, measurement tools, and time between measurements for each movement parameter.

TABLE 1

CHARACTERISTICS FOR EACH MOVEMENT TASK

Movement Category	Movement Parameter/ Measurement Type	Sample Size, n	Measurement Tools (n Studies)	Pain/Activity Limitation Measurement Tools (n Studies)	Time Between Measurements, wk
Flexion	Flexion ROM (peak)				
	Lumbar ^{6,10,19,23,40,47,59,62,66}	Range, 21-456; median, 89; IQR, 79.5 ^a	Inclinometer or goniometer (5), electromechanical (2), motion sensor (1), Schober test (1)	Pain: VAS (2), NPRS (2) Activity limitation: RMDQ (2), ODI (3), PSFS (1), QBPDS (1), HFQ (1), LBP Disability Index (1)	Range, 3-12; median, 6.5; IQR, 5 ^a
	Multisegment ^{9-11,38,54,60,62,66}	Range, 17-111; median, 38.5; IQR, 41.2 ^a	Fingertip to floor (4), motion sensor (2), electromechanical (1), rachimetry (1)	Pain: NPRS (2), VAS (1) Activity limitation: RMDQ (4), ODI (3), HFQ (1)	Range, 4-38; median, 9; IQR, 4.5 ^a
	Flexion sEMG variables				
	Lumbar ^{41,59,61,62,69,73}	Range, 18-68; median, 34.5; IQR, 24 ^a	sEMG (flexion relaxation of lumbar paraspinal muscles) (6)	Pain: NPRS (2), VAS (3) Activity limitation: ODI (3), RMDQ (1)	Range, 1-12; median, 5.5; IQR, 4 ^a
Extension	Flexion velocity and rhythm				
	Lumbar ⁶²	33 (only 1 study)	Motion sensor	Did not measure pain Activity limitation: ODI	8 (only 1 study)
	Multisegment ^{54,60,62}	Range, 33-111; median, 44; IQR, 78 ^a	Motion sensor (2), electromechanical (1)	Pain: NRS (1) Activity limitation: ODI (3)	Range, 8-38; median, 12; IQR, 30 ^a
Lateral flexion	Extension ROM (peak)				
	Lumbar ^{6,19,23,47}	Range, 25-456; median, 89; IQR, 109.25 ^a	Inclinometer (4)	Pain: VAS (2) Activity limitation: ODI (1), PSFS (1), QBPDS (1), LBP Disability Index (1)	Range, 3-11; median, 5.5; IQR, 5.75 ^a
Rotation	Lateral flexion ROM (peak)				
	Lumbar ^{12,47}	Range, 56-456; median, 256; IQR, 400 ^a	Inclinometer (1), electromechanical (1)	Pain: MPQ (1) Activity limitation: LBP Disability index (1)	Range, 2-3; median, 2.5; IQR, 1 ^a
Summative	Multisegment ^{2,23,66}	Range, 88-98; median, 96; IQR, 10 ^a	Fingertip to floor (3)	Pain: NRS (1) Activity limitation: RMDQ (2), ODI (1)	Range, 3-3.5; median, 3.5; IQR, 0.5 ^a
	Rotation ROM (peak)				
	Lumbar ^{12,23}	Range, 56-88; median, 72; IQR, 32 ^a	Electromechanical (1), goniometer (1)	Pain: modified MPQ (1) Activity limitation: ODI (1)	Range, 2-3; median, 2.5; IQR, 1 ^a
Other	Multisegment ⁴⁷	456 participants	Compass	Did not measure pain Activity limitation: LBP Disability Index	3
	Summative ROM (peaks)				
	Lumbar ^{12,39,42,47,49}	Range, 36-456; median, 132; IQR, 143 ^a	Electromechanical (3), inclinometer (1), vector stereograph (1)	Pain: NRS (2), modified MPQ (1) Activity limitation: RMDQ (2), LBP Disability Index (1), 12-item functional impairment VAS (1)	Range, 2-12; median, 4; IQR, 9 ^a
Phase-plot area for spinal kinematics during lifting	Multisegment ^{24,45,46}	Range, 27-476; median, 194; IQR, 449 ^a	Schober test/fingertip to floor/compass (2), inclinometer (1)	Pain: 5-point NRS (current and average over 1 wk) (1) Activity limitation: FCI (1), LBP Disability Index (1), 12-item functional impairment scale (1)	Range, 20-72; median, 52; IQR, 52 ^a
	Multisegment ^{35,49}	Range, 112-199; median, 155.5; IQR, 87 ^a	Electromechanical (2)	Pain: NRS (1) Activity limitation: ODI (1), RMDQ (1)	Range, 12-29.3; median, 20.7; IQR, 17.3 ^a

Abbreviations: FCI, Functional Capacity Index; HFQ, Hannover Functional Questionnaire; IQR, interquartile range; LBP, low back pain; MPQ, McGill Pain Questionnaire; NPRS, numeric pain-rating scale; NRS, numeric rating scale; ODI, Oswestry Disability Index; PSFS, Patient-Specific Functional Scale; QBPDS, Quebec Back Pain Disability Scale; RMDQ, Roland-Morris Disability Questionnaire; ROM, range of motion; sEMG, surface electromyography; VAS, visual analog scale.

^aThe IQR was calculated as the third quartile minus the first quartile.

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The 27 included studies investigated the relationship between movement and pain or movement and activity limitation at the individual level 65 times (pain, 26 times; activity limitation, 39 times) (TABLE 2). Movements related to flexion (ROM, velocity, electromyography, and lumbopelvic rhythm) accounted for 57% (37/65) of all movements measured. The majority (23/37, 62%) of the flexion parameters were related to flexion ROM. Parameters related to extension were measured 8% (5/65) of the time, lateral flexion 9% (6/65) of the time, rotation 5% (3/65) of the time, summative ROM in the cardinal planes 17% (11/65) of the time, and other measures (such as phase-plot area or lifting spinal position) 5% (3/65) of the time. No studies reporting the lumbar contribution to flexion or timing of lumbar kinematics were identified.

Quality Assessment Within Studies

The risk-of-bias scores at individual study and domain levels are presented

in TABLE 3, along with their categorization (low, medium, or high risk of bias). All 27 included studies adequately described the participant demographic and clinical characteristics, 24 (89%) adequately described the selection criteria, 19 (70%) had acceptable missingness of data, all reported measuring movements in standardized and valid ways, 26 (96%) reported using reliable measures of movement, 25 (93%) reported using standardized, valid, and reliable pain or activity limitation questionnaires, all (100%) used appropriate statistical analysis, and 3 (11%) reported using an assessor blind to movement, pain, or activity limitation outcome.

We categorized 25 of the 27 studies (93%) as having a high risk of bias, mostly because of uncertainty about assessor blinding, which was only reported in 3 studies (it is possible that some studies blinded assessors but did not report it). Authorities (such as the Cochrane Collaboration⁶⁵) advise that when the outcome assessor is not blinded and an outcome

(in this case, movement assessment) could have been influenced by knowledge of the intervention received (analogous here to knowledge of change in pain or activity limitation), review authors should assess whether it is likely that such influence occurred. In our judgment, there was a high risk that lack of blinding would affect the measurement of movement.

Quality Assessment Across Studies

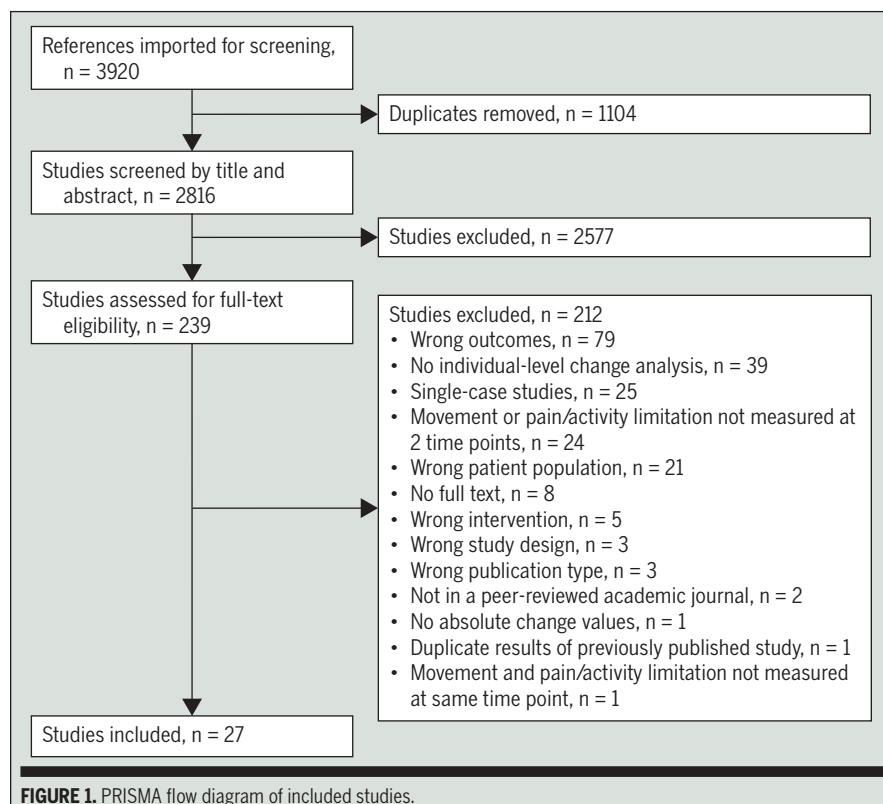
Starting with a GRADE score of “medium quality” for cohort-derived data,^{5,18,63} after consideration of all GRADE criteria, we downgraded the classification to low-quality evidence, defined by GRADE⁵ as, “Our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.” A full description of our GRADE assessment across all domains is detailed in APPENDIX C.

Synthesis of Results

The summary results about the relationship between changes in movement parameters, pain, and activity limitation, and the direction of this relationship when present, are displayed in TABLE 2.

Movement and Pain A relationship between a change in movement and a change in pain was investigated 26 times in 16 studies^{2,6,10,12,19,24,38,40,42,49,54,59,61,62,69,73} (some studies investigated multiple movements), with a relationship found 22% (5.6/26) of the time. In studies that found a relationship,^{2,12,40,49,61} increased movement range, velocity, or flexion relaxation was associated with improved pain 100% (5.6/5.6) of the time. The most common measures of pain were the numeric pain-rating scale and the 100-mm visual analog scale (both used 6 times), with other measures of pain (eg, the McGill Pain Questionnaire) used 3 times.

Movement and Activity Limitation A relationship between a change in movement and a change in activity limitation was investigated 39 times in 23 studies^{2,9,11,19,23,24,35,38-42,45-47,49,54,59,60,62,66,69,73} (some studies investigated multiple movements), with a relation-



ship found 37% (14.4/39) of the time. In the studies that found a relationship,^{9,11,23,38,40,41,45-47,49,54,60,62,66} increased movement range, velocity, or flexion relaxation was associated with improved activity limitation 90% (12.9/14.4) of the time. The most common measure of activity limitation was the Roland-Morris Disability Questionnaire (used 10 times), followed by the Oswestry Disability Index (used 5 times), with other measures (such as the Quebec Back Pain Disability Scale, the Functional Capacity Index, and the Hannover Functional Questionnaire) used 7 times.

Movement, Outcome, and Direction When considering investigations of pain and activity limitation together, a relationship between movement and

outcome was observed 31% (20.0/65.0) of the time. Increased range, velocity, or flexion relaxation was associated with improved clinical outcome 93% (18.5/20.0) of the time.

Sensitivity Analysis

As 25 of 27 (93%) studies were categorized as having a high risk of bias, there were insufficient data to perform a sensitivity analysis of the prevalence of a relationship in only the studies with a low risk of bias.

Baseline values of pain and activity limitation in the included studies were comparable to the values reported in NSLBP RCTs,^{21,25,68} above reported levels for “disabling back pain,”⁷¹ and similar between the studies that did and did not

identify a relationship. The detailed results of these additional analyses can be found in **APPENDIX E**.

Two video abstracts of this systematic review are also available (see <https://youtu.be/iUGVOdtQPis> for the full video abstract and <https://youtu.be/yFesgTl-ZynM> for the brief video abstract).

DISCUSSION

THERE WAS A RELATIONSHIP BETWEEN changes in volitional spinal movement (including muscle activity) and changes in pain or activity limitation 31% of the time (20 of the 65 associations tested) in 27 low-quality studies involving 2739 participants. The relationship was between increased

TABLE 2

SUMMARY OF RESULTS ABOUT A RELATIONSHIP BETWEEN CHANGE IN MOVEMENT AND CHANGE IN PAIN OR ACTIVITY LIMITATION

Movement Category	Movement Parameter/ Measurement Type	Movement and Pain		Movement and Activity Limitation	
		Times Investigated, n	Supporting Relationship, n (direction) ^a	Times Investigated, n	Supporting Relationship, n (direction) ^a
Flexion	Flexion ROM				
	Lumbar	Five ^{6,13,40,59,62}	1 (1↑, 0↓)	Seven ^{19,23,40,47,59,62,66}	2.5 (1.5↑, 1↓)
	Multisegment	Four ^{10,40,54,62}	0	Seven ^{11,38,54,60,62,66}	5 (5↑, 0↓)
	EMG (FRR) ^b	Five ^{59,61,62,69,73}	0.25 (0.25↑, 0↓)	Four ^{41,59,62,73}	1 (1↑, 0↓)
Extension	Velocity and rhythm	Two ^{54,62}	0	Three ^{54,60,62}	0.1 (0.1↑, 0↓)
	Extension range				
Lateral flexion	Lumbar	Two ^{6,19}	0	Three ^{19,23,47}	0
	Lateral flexion ROM				
Rotation	Lumbar	One ¹²	1 (1↑, 0↓)	One ⁴⁷	1 (1↑, 0↓)
	Multisegment	One ²	1 (1↑, 0↓)	Three ^{2,23,66}	1.5 (1.5↑, 0↓)
Summative	Rotation ROM				
	Lumbar	One ¹²	1 (1↑, 0↓)	One ²³	1 (1↑, 0↓)
Other	Multisegment	...	...	One ⁴⁷	1 (1↑, 0↓)
	Summative ROM				
Other	Lumbar	Three ^{12,42,49}	1.3 (1.3↑, 0↓)	Four ^{39,42,47,49}	0.3 (0.3↑, 0↓)
	Multisegment	One ²⁴	0	Three ^{24,45,46}	1 (0.5↑, 0.5↓)
Other	Other ^c				
	Lumbar	One ⁴⁹	0	Two ^{35,49}	0

Abbreviations: EMG, electromyography; FRR, flexion-relaxation ratio; ROM, range of motion.

^aVote-counting system (scored 1.0 per parameter). For studies that used multiple measures for a single movement parameter, the value of 1 was divided by the total number of measures used. For example, if a study only used an inclinometer to measure flexion ROM, and that parameter showed a relationship, then the vote from that study was 1.0. However, if that study also utilized 3 other methods of flexion ROM measurement, and only the inclinometer measurement showed a relationship, then the vote from that study would be 0.25. The ↑ symbol means that an increase in that parameter was related to an improvement in pain or activity limitation, and the ↓ symbol means that a decrease in that parameter was related to an improvement in pain or activity limitation.

^bA higher value indicates greater relaxation (less muscle activity) during peak flexion.

^cIncluded phase-plot area and spinal kinematics during lifting.

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range, velocity, or flexion relaxation and improved pain and activity limitation 93% of the time (18.5 of 20 relationships observed) (FIGURE 2). Baseline pain and activity limitation did not appear to explain the presence of a relationship.

Quality of the Evidence

The low overall quality of evidence (by GRADE assessment criteria) was influenced predominantly by 2 domains:

domain 1 (risk of bias) and domain 4 (indirectness). A lack of reporting on assessor blinding influenced the risk of bias domain, and designs and interventions that could not accommodate the heterogeneous movements in people with NSLBP³⁷ influenced the indirectness domain.

Interpretation of Findings

The infrequency of a relationship may be evidence that, in most people with

NSLBP, changes in pain or activity limitation are unrelated to changes in movement. Alternatively, it could be evidence that some people with NSLBP move normally and not all of them have a movement problem that is amenable to change. A recent cohort study of 266 people found that a subgroup (25%) of people with NSLBP move normally, despite their pain and activity limitation.³⁰ Further, our findings could be evidence

TABLE 3

RISK-OF-BIAS ASSESSMENT^a

Study	Item ^b									Overall Risk ^d
	1	2	3	4 ^c	5 ^c	6 ^c	7 ^c	8 ^c	9 ^c	
Anderson et al ²	1	1	1	1	1	1	1	1	U	High
Beladev and Masharawi ⁶	1	1	1	1	1	1	1	1	U	High
Caporaso et al ⁹	1	1	1	1	1	1	1	1	U	High
Demoulin et al ¹⁰	1	1	0	1	1	1	1	1	U	High
Ekedahl et al ¹¹	1	1	0	1	1	1	1	1	U	High
Elnaggar et al ¹²	1	0	1	1	1	1	1	1	U	High
Helmhout et al ¹⁹	1	1	0	1	1	1	1	1	U	High
Hurri et al ²³	1	1	U	1	1	1	1	1	U	High
Johannsen et al ²⁴	1	1	0	1	1	1	1	1	0	High
Loisel et al ³⁵	1	1	1	1	U	1	1	1	1	High
Mannion et al ⁴⁰	1	1	1	1	1	1	1	1	U	High
Mannion et al ³⁹	1	1	1	1	1	1	1	1	U	High
Mannion et al ³⁸	1	1	1	1	1	1	1	1	1	Low
Marshall and Murphy ⁴¹	1	1	1	1	1	1	1	1	U	High
Martin et al ⁴²	1	1	0	1	1	U	U	1	U	High
Mellin et al ⁴⁷	1	1	1	1	1	1	1	1	U	High
Mellin et al ⁴⁵	1	1	1	1	1	1	1	1	U	High
Mellin et al ⁴⁶	1	0	0	1	1	U	U	1	U	High
Mieritz et al ⁴⁹	1	1	0	1	1	1	1	1	U	High
Nordstoga et al ⁵⁴	1	1	1	1	1	1	1	1	U	High
Pagé et al ⁵⁹	1	1	1	1	1	1	1	1	U	High
Poitras et al ⁶⁰	1	1	1	1	1	1	1	1	1	Low
Qiao et al ⁶¹	1	1	1	1	1	1	1	1	U	High
Shahvarpour et al ⁶²	1	1	1	1	1	1	1	1	U	High
Strand et al ⁶⁶	1	0	1	1	1	1	1	1	U	High
Watson et al ⁶⁹	1	1	1	1	1	1	1	1	U	High
Xia et al ⁷³	1	1	1	1	1	1	1	1	U	High
Domain-level prevalence	100%	89%	70%	100%	96%	93%	93%	100%	11%	...

Abbreviation: U, unclear.

^aModified from Moola et al.⁵¹ Scoring: 1, adequate; 0, not adequate.

^bItems: 1, Demographics; 2, Selection criteria; 3, Missingness; 4, Movement standard/valid; 5, Movement reliable; 6, Pain/activity limitation standard/valid; 7, Pain/activity limitation reliable; 8, Appropriate analysis; 9, Assessor blinding.

^cCritical risk of bias.

^dHigh risk, any of the critical risks were scored negatively or were unclear; medium risk, all of the critical risks were scored positively, but some noncritical risks were scored negatively or were unclear; low risk, scored positively on all items.

that an unmeasured factor was the main catalyst for change, and this factor may differ among individuals.

In previous research, improvements in psychological factors (eg, distress, pain catastrophizing, or fear avoidance) were more strongly related to improved pain or activity limitation than to movement.^{38,39,54} Positive experiences and reduced sense of threat following the targeting and safe completion of movements perceived as dangerous or damaging may improve pain and activity limitation, irrespective of change in the targeted movements. Therefore, some therapeutic effects observed with interventions targeting movement may not be related to change in that movement, but simply to changes in the perceived threat value of movement.

Factors That May Influence the Relationship Between Change in Movement and Change in Pain or Activity Limitation

While it is possible that no relationship between changes in movement and changes in pain or activity limitation may exist at the individual level, several scenarios could preclude the identification of such a relationship. Due to the heterogeneity of NSLBP,^{37,56} the same factors may not contribute to participants' experience of pain and activity limitation. Specifically, some participants may not have an impairment of the movement measured. Not measuring individually relevant movement parameters is a potential limitation among cohort studies that rely on consistent parameters measured across all participants. For example, if flexion is the measured movement parameter among a cohort of people with NSLBP and some did not have flexion limitations but reported limitations in extension, only analyzing relationships between changes in flexion and changes in pain or activity limitation would underestimate the true relationship between changes in movement and changes in pain or activity limitation. The lack of a frequent relationship in our systematic

review of cohort studies may indicate that related movement parameters are person specific.

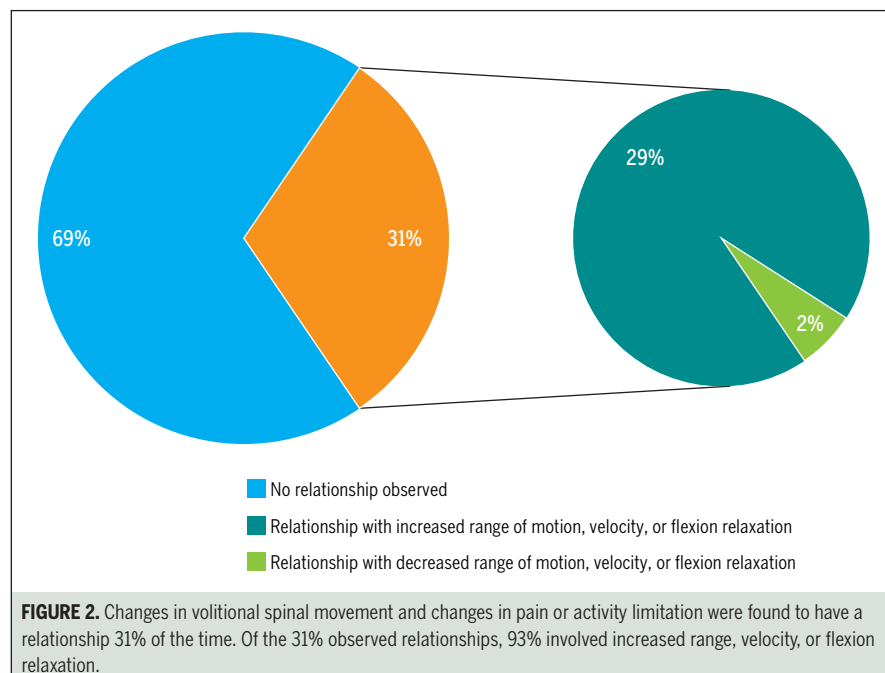
Frequently, a movement parameter measured and compared to changes in pain or activity limitation will not be the one directly targeted by the intervention. When the measured movement and aims of the intervention are not well aligned, the prevalence of relationships between changes in movement and changes in pain may be underestimated. For example, where studies^{6,9,11,24,38 40,49,62} have measured maximum flexion ROM or flexion relaxation as an outcome and found a relationship between its increase and an improvement in the condition, the intervention encouraged movement restriction or stabilization of the spine, neutral positions, and bracing of the trunk muscles, which might have limited changes in the measured movement parameter, precluding the identification of a relationship. Additionally, for example, heterogeneity across the duration of participants' back pain (acute through persistent) and their backgrounds (age, occupation, working versus not working, athletes versus non-athletes) might have influenced the identification of a relationship.

There was substantial heterogeneity in the measurement of movement, pain and activity limitation, and the analysis of change in the included studies. The selection of the measured movement parameter (whether it was salient for the participants), the validity and reliability of the tool used to measure movement, as well as the type, reliability, and validity of the pain or activity limitation measure might have influenced our results. However, the risk of bias assessment judged the validity and reliability of the movement, pain, and activity limitation measures to be acceptable, and therefore this is unlikely to have influenced our findings.

Finally, there was considerable variability in the time interval between measures. Some studies measured changes within a week,⁶¹ while others measured changes over a 1.5-year period.⁴⁵

Changes in Movement When a Relationship Was Found

When a relationship was identified, the results regarding the direction of movement changes that were related to improved pain or activity limitation were very consistent. Improved pain or activity



limitation was consistently associated with increased spinal movement range, velocity, or flexion relaxation of the back extensor muscles. Improvement may represent a return toward normal movement, as there is considerable evidence that people without LBP move, on average, with greater range, velocity, and flexion relaxation than people with LBP.²⁹⁻³¹ When changes in movement are related to improved pain or activity limitation, it is predominantly in recovery from a protective movement behavior. However, the causal direction remains unknown (whether reducing protective movement improves pain and activity limitation or improving pain and activity limitation reduces protective movement).

Psychological factors, such as fear of movement and self-efficacy, have been associated with a protective pattern of movement,^{27,43,55,58} as well as pain and activity limitation outcomes.^{1,38,39,54} It is plausible that reducing protective movement improves pain or activity limitation, a change potentially mediated by a reduction in pronociceptive protective behaviors (muscle guarding and movement restriction)⁵⁷ or improvements in psychological factors, such as fear of movement and self-efficacy.^{8,33} Conversely, improvements in pain or activity limitation due to phenomena such as natural history, placebo, nonspecific effects, or changes in other biopsychosocial factors^{4,48,72} may result in reducing protective movement.

To our knowledge, this is the first published study that has investigated the relationship between changes across comprehensive volitional movement parameters and changes in pain and activity limitation at the individual level. The only other systematic review with individual-level analysis on this topic only considered spinal ROM parameters as measures of volitional spinal movement.⁶⁴ We also considered velocity, spinal muscle activity, extension ROM, ROM during functional tasks, lumbar contribution to flexion, and the timing of lumbar kinematics. We found a prevalence of 31% (20 of 65 associations),

similar to the previous review's finding of 34% (6.5/19 associations),⁶⁴ despite including broader movement parameters, which increases our confidence in the results. The results of our systematic review are also comparable to those of a previous systematic review³² in which associations between changes in movement and improvement in pain or activity limitation were infrequently observed at the group level.

Strengths

The broad selection criteria were a strength of this systematic review. Including numerous study designs, all physical interventions, all durations of NSLBP, various volitional movement measures, and all studies since database inception; searching multiple electronic databases; and using a comprehensive search strategy developed in collaboration with a faculty librarian mean that it is likely that we captured the pertinent studies.

Limitations

Potential limitations are language bias (due to only including English-language publications), publication bias (due to only including peer-reviewed studies), selective reporting by the authors of included studies (due to a lack of registered study protocols), exclusion of studies that used surgical or pharmacological interventions, and the use of a conventional ($P < .05$) but arbitrary threshold of statistical significance when identifying the presence of a relationship. Additionally, due to the heterogeneity of the included studies, vote counting was the most appropriate method to synthesize the results.

Considerations

Changes From PROSPERO Registration Following inspection of the size of the initial search yield, we restricted the scope of the systematic review to (1) include only movements and exclude static postures, (2) exclude surgical or pharmacological interventions, and (3) exclude low-quality single-case studies. The

PROSPERO registration was therefore reworded to reflect those changes.

Clinical Implications

Infrequent identification of a relationship between changes in movement and pain suggests that movement does not necessarily have to change for pain or activity limitation to improve in people with NSLBP. However, a lack of high-quality evidence limits our confidence in this implication. The movement parameters that were consistently associated with improved pain or activity limitation (increased spinal range, velocity, and flexion relaxation) can be observed, targeted, and monitored by clinicians. These movement parameters may have different relevance for different people and may be influenced by various biopsychosocial factors such as levels of fear and self-efficacy,^{1,27,43,55,58} which may also require observation, targeting, and monitoring. The lack of a clear relationship lends support to calls for a multidimensional, individualized biopsychosocial approach to managing NSLBP.

Unanswered Questions and Future Research

Much remains unknown about which changes in physical, cognitive, emotional, social, lifestyle, or other factors are important in order to see a subsequent improvement in outcome. High-quality studies that investigate relationships between changes in individually relevant biopsychosocial factors and pain or activity limitation at the individual level would increase our understanding of the factors related to improved outcomes. The quality of future studies could be improved by reporting on assessor blinding, accommodating and assessing movements specific to the limitations of the individual, and using multidimensional interventions that target those individually relevant movements and limitations.

There is little research investigating the relationship between changes in individually relevant movement and changes in pain or activity limitation, or the or-

der in which these occur. Study designs that accommodate heterogeneity and repeatedly measure numerous multidimensional factors, including individually relevant movement, such as rigorous single-case designs^{7,15} as well as mediation analyses of large-cohort and RCT data may help provide insights into causality²⁰ and mechanisms of improvement among people with NSLBP. Despite the infrequent identification of a relationship in the current literature, the existing evidence base is predominantly of low quality. Thus, high-quality research is required to provide more dependable insights about this topic.

CONCLUSION

A CHANGE IN VOLITIONAL SPINAL movement was infrequently (31%, or 20 of the 65 associations investigated) related to a concomitant change in pain or activity limitation in people with NSLBP. This was observed across multiple movement parameters, measures, study periods, and interventions. When a change in movement was associated with improved pain or activity limitation, it almost always occurred via an increase in spinal range, velocity, or flexion relaxation of the back extensor muscles. ●

KEY POINTS

FINDINGS: There was low-quality evidence that changes in volitional spinal movement (including muscle activity) were infrequently (31%, or 20/65 associations tested) related to changes in pain or activity limitation at the individual level among people with nonspecific low back pain. Where relationships were observed, improved pain and activity limitation were consistently (93%, or 18.5/20 relationships observed) related to increased spinal movement range, velocity, or flexion relaxation of spinal extensor muscles.

IMPLICATIONS: Infrequent observation of a relationship between changes in these domains suggests that movement does not necessarily need to change for

pain or activity limitation to improve. Changes in other biopsychosocial factors may relate more to improvement in low back pain, and these may be different for different people, lending support to calls for a multidimensional and individualized approach to the assessment and management of nonspecific low back pain.

CAUTION: Considerable heterogeneity among studies, numerous methodological limitations, and a paucity of high-quality evidence preclude a definitive understanding of these relationships.

STUDY DETAILS

AUTHOR CONTRIBUTIONS: All authors contributed to the conception, design, analysis, and interpretation of this study.

All authors contributed to drafting and critically revising the manuscript and assume responsibility for the work.

DATA SHARING: All data relevant to the manuscript are included in the article or within the appendices.

PATIENT AND PUBLIC INVOLVEMENT: There was no patient or public involvement in this research.

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REFERENCES

1. Alhewimel A, Alotaibi M, Radford K, Coulson N. Psychosocial factors associated with change in pain and disability outcomes in chronic low back pain patients treated by physiotherapist: a systematic review. *SAGE Open Med*. 2018;6:2050312118757387. <https://doi.org/10.1177/2050312118757387>
2. Anderson B, Lygren H, Magnussen LH, Eide GE, Strand LI. What functional aspects explain patients' impression of change after rehabilitation for long-lasting low back pain? *Physiother Res Int*. 2013;18:167-177. <https://doi.org/10.1002/pri.1548>
3. Arthritis and Osteoporosis Victoria. A Problem Worth Solving. Elsternwick, Australia: Arthritis and Osteoporosis Victoria; 2013.
4. Artus M, van der Windt DA, Jordan KP, Hay EM. Low back pain symptoms show a similar pattern of improvement following a wide range of primary care treatments: a systematic review of randomized clinical trials. *Rheumatology (Oxford)*. 2010;49:2346-2356. <https://doi.org/10.1093/rheumatology/keq245>
5. Balshem H, Helfand M, Schünemann HJ, et al. GRADE guidelines: 3. Rating the quality of evidence. *J Clin Epidemiol*. 2011;64:401-406. <https://doi.org/10.1016/j.jclinepi.2010.07.015>
6. Beladev N, Masharawi Y. The effect of group-exercising on females with non-specific chronic low back pain in a sitting position: a pilot study. *J Back Musculoskeletal Rehabil*. 2011;24:181-188. <https://doi.org/10.3233/BMR-2011-0292>
7. Borckardt JJ, Nash MR, Murphy MD, Moore M, Shaw D, O'Neil P. Clinical practice as natural laboratory for psychotherapy research: a guide to case-based time-series analysis. *Am Psychol*. 2008;63:77-95. <https://doi.org/10.1037/0003-066X.63.2.77>
8. Caneiro JP, Smith A, Linton SJ, Moseley GL, O'Sullivan P. How does change unfold? An evaluation of the process of change in four people with chronic low back pain and high pain-related fear managed with Cognitive Functional Therapy: a replicated single-case experimental design study. *Behav Res Ther*. 2019;117:28-39. <https://doi.org/10.1016/j.brat.2019.02.007>
9. Caporaso F, Pulkovski N, Sprott H, Mannion AF. How well do observed functional limitations explain the variance in Roland-Morris scores in patients with chronic non-specific low back pain undergoing physiotherapy? *Eur Spine J*. 2012;21 suppl 2:S187-S195. <https://doi.org/10.1007/s00586-012-2255-6>
10. Demoulin C, Maquet D, Tomasella M, Croisier JL, Crielaard JM, Vanderthommen M. Benefits of a physical training program after back school for chronic low back pain patients. *J Musculoskeletal Pain*. 2006;14:21-31. https://doi.org/10.1300/J094v14n02_04
11. Ekedahl H, Jönsson B, Frobell RB. Fingertip-to-floor test and straight leg raising test: validity, responsiveness, and predictive value in patients with acute/subacute low back pain. *Arch Phys Med Rehabil*. 2012;93:2210-2215. <https://doi.org/10.1016/j.apmr.2012.04.020>
12. Elnaggar IM, Nordin M, Sheikhzadeh A, Parnianpour M, Kahanovitz N. Effects of spinal flexion and extension exercises on low-back pain and spinal mobility in chronic mechanical low-back pain patients. *Spine (Phila Pa 1976)*. 1991;16:967-972. <https://doi.org/10.1097/00007632-199108000-00018>
13. Falla D, Gizzi L, Tschapek M, Erlenwein J, Petzke F. Reduced task-induced variations in the distribution of activity across back muscle regions in individuals with low back pain. *Pain*. 2014;155:944-953. <https://doi.org/10.1016/j.pain.2014.01.027>
14. Frymoyer JW. Back pain and sciatica. *N Engl J Med*. 1988;318:291-300. <https://doi.org/10.1056/NEJM198802043180506>
15. Gaynor ST, Harris A. Single-participant assessment of treatment mediators: strategy description and examples from a behavioral

activation intervention for depressed adolescents. *Behav Modif.* 2008;32:372-402. <https://doi.org/10.1177/0145445507309028>

16. GBD 2015 DALYs and HALE Collaborators. Global, regional, and national disability-adjusted life-years (DALYs) for 315 diseases and injuries and healthy life expectancy (HALE), 1990–2015: a systematic analysis for the Global Burden of Disease Study 2015. *Lancet.* 2016;388:1603-1658. [https://doi.org/10.1016/S0140-6736\(16\)31460-X](https://doi.org/10.1016/S0140-6736(16)31460-X)
17. GBD 2015 Disease and Injury Incidence and Prevalence Collaborators. Global, regional, and national incidence, prevalence, and years lived with disability for 310 diseases and injuries, 1990–2015: a systematic analysis for the Global Burden of Disease Study 2015. *Lancet.* 2016;388:1545-1602. [https://doi.org/10.1016/S0140-6736\(16\)31678-6](https://doi.org/10.1016/S0140-6736(16)31678-6)
18. Guyatt GH, Oxman AD, Vist GE, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. *BMJ.* 2008;336:924-926. <https://doi.org/10.1136/bmj.39489.470347AD>
19. Helmhout PH, Witjes M, Nijhuis-van der Sanden RW, Bron C, van Aalst M, Staal JB. The effects of lumbar extensor strength on disability and mobility in patients with persistent low back pain. *J Sports Med Phys Fitness.* 2017;57:411-417. <https://doi.org/10.2376/S0022-4707.16.06270-8>
20. Hill AB. The environment and disease: association or causation? *Proc R Soc Med.* 1965;58:295-300. <https://doi.org/10.1177/003591576505800503>
21. Hill JC, Whitehurst DG, Lewis M, et al. Comparison of stratified primary care management for low back pain with current best practice (STaRT Back): a randomised controlled trial. *Lancet.* 2011;378:1560-1571. [https://doi.org/10.1016/S0140-6736\(11\)60937-9](https://doi.org/10.1016/S0140-6736(11)60937-9)
22. Hodges PW, Richardson CA. Altered trunk muscle recruitment in people with low back pain with upper limb movement at different speeds. *Arch Phys Med Rehabil.* 1999;80:1005-1012. [https://doi.org/10.1016/S0003-9993\(99\)90052-7](https://doi.org/10.1016/S0003-9993(99)90052-7)
23. Hurri H, Huupli M, Alaranta H. Strength and mobility measurements in assessing the outcome of an intensive back rehabilitation program. *Eur J Phys Med Rehabil.* 1995;5:160-164.
24. Johannsen F, Remvig L, Kryger P, et al. Exercises for chronic low back pain: a clinical trial. *J Orthop Sports Phys Ther.* 1995;22:52-59. <https://doi.org/10.2519/jospt.1995.22.2.52>
25. Kamper SJ, Apeldoorn AT, Chiarotto A, et al. Multidisciplinary biopsychosocial rehabilitation for chronic low back pain. *Cochrane Database Syst Rev.* 2014;CD000963. <https://doi.org/10.1002/14651858.CD000963.pub3>
26. Karayannis NV, Jull GA, Hodges PW. Movement-based subgrouping in low back pain: synergy and divergence in approaches. *Physiotherapy.* 2016;102:159-169. <https://doi.org/10.1016/j.physio.2015.04.005>
27. Karayannis NV, Smeets RJ, van den Hoorn W,

Hodges PW. Fear of movement is related to trunk stiffness in low back pain. *PLoS One.* 2013;8:e67779. <https://doi.org/10.1371/journal.pone.0067779>

28. Koes BW, van Tulder M, Lin CW, Macedo LG, McAuley J, Maher C. An updated overview of clinical guidelines for the management of non-specific low back pain in primary care. *Eur Spine J.* 2010;19:2075-2094. <https://doi.org/10.1007/s00586-010-1502-y>
29. Laird RA, Gilbert J, Kent P, Keating JL. Comparing lumbo-pelvic kinematics in people with and without back pain: a systematic review and meta-analysis. *BMC Musculoskelet Disord.* 2014;15:229. <https://doi.org/10.1186/1471-2474-15-229>
30. Laird RA, Keating JL, Kent P. Subgroups of lumbo-pelvic flexion kinematics are present in people with and without persistent low back pain. *BMC Musculoskelet Disord.* 2018;19:309. <https://doi.org/10.1186/s12891-018-2233-1>
31. Laird RA, Keating JL, Ussing K, Li P, Kent P. Does movement matter in people with back pain? Investigating 'atypical' lumbo-pelvic kinematics in people with and without back pain using wireless movement sensors. *BMC Musculoskelet Disord.* 2019;20:28. <https://doi.org/10.1186/s12891-018-2387-x>
32. Laird RA, Kent P, Keating JL. Modifying patterns of movement in people with low back pain – does it help? A systematic review. *BMC Musculoskelet Disord.* 2012;13:169. <https://doi.org/10.1186/1471-2474-13-169>
33. Lee H, Mansell G, McAuley JH, et al. Causal mechanisms in the clinical course and treatment of back pain. *Best Pract Res Clin Rheumatol.* 2016;30:1074-1083. <https://doi.org/10.1016/j.berh.2017.04.001>
34. Lin IB, O'Sullivan PB, Coffin JA, Mak DB, Toussaint S, Straker LM. Disabling chronic low back pain as an iatrogenic disorder: a qualitative study in Aboriginal Australians. *BMJ Open.* 2013;3:e002654. <https://doi.org/10.1136/bmjopen-2013-002654>
35. Loisel P, Poitras S, Lemaire J, Durand P, Southière A, Abenhaim L. Is work status of low back pain patients best described by an automated device or by a questionnaire? *Spine (Phila Pa 1976).* 1998;23:1588-1594; discussion 1595. <https://doi.org/10.1097/00007632-199807150-00015>
36. Macrae IF, Wright V. Measurement of back movement. *Ann Rheum Dis.* 1969;28:584-589. <https://doi.org/10.1136/ard.28.6.584>
37. Maher C, Underwood M, Buchbinder R. Non-specific low back pain. *Lancet.* 2017;389:736-747. [https://doi.org/10.1016/S0140-6736\(16\)30970-9](https://doi.org/10.1016/S0140-6736(16)30970-9)
38. Mannion AF, Caporaso F, Pulkovski N, Spott H. Spine stabilisation exercises in the treatment of chronic low back pain: a good clinical outcome is not associated with improved abdominal muscle function. *Eur Spine J.* 2012;21:1301-1310. <https://doi.org/10.1007/s00586-012-2155-9>
39. Mannion AF, Junge A, Taimela S, Muntener M, Lorenzo K, Dvorak J. Active therapy for chronic

low back pain: part 3. Factors influencing self-rated disability and its change following therapy. *Spine (Phila Pa 1976).* 2001;26:920-929. <https://doi.org/10.1097/00007632-200104150-00015>

40. Mannion AF, Muntener M, Taimela S, Dvorak J. A randomized clinical trial of three active therapies for chronic low back pain. *Spine (Phila Pa 1976).* 1999;24:2435-2448. <https://doi.org/10.1097/00007632-199912010-00004>
41. Marshall PW, Murphy BA. Evaluation of functional and neuromuscular changes after exercise rehabilitation for low back pain using a Swiss ball: a pilot study. *J Manipulative Physiol Ther.* 2006;29:550-560. <https://doi.org/10.1016/j.jmpt.2006.06.025>
42. Martin PR, Rose MJ, Nichols PJ, Russell PL, Hughes IG. Physiotherapy exercises for low back pain: process and clinical outcome. *Int Rehabil Med.* 1986;8:34-38. <https://doi.org/10.3109/03790798609166509>
43. Matheve T, De Baets L, Bogaerts K, Timmermans A. Lumbar range of motion in chronic low back pain is predicted by task-specific, but not by general measures of pain-related fear. *Eur J Pain.* 2019;23:1171-1184. <https://doi.org/10.1002/ejp.1384>
44. McGorry RW, Lin JH. Flexion relaxation and its relation to pain and function over the duration of a back pain episode. *PLoS One.* 2012;7:e39207. <https://doi.org/10.1371/journal.pone.0039207>
45. Mellin G, Härkäpää K, Hurri H, Järvinen A. A controlled study on the outcome of inpatient and outpatient treatment of low back pain. Part IV. Long-term effects on physical measurements. *Scand J Rehabil Med.* 1990;22:189-194.
46. Mellin G, Härkäpää K, Vanharanta H, Huupli M, Heinonen R, Järvinen A. Outcome of a multimodal treatment including intensive physical training of patients with chronic low back pain. *Spine (Phila Pa 1976).* 1993;18:825-829. <https://doi.org/10.1097/00007632-199306000-00003>
47. Mellin G, Hurri H, Härkäpää K, Järvinen A. A controlled study on the outcome of inpatient and outpatient treatment of low back pain. Part II. Effects on physical measurements three months after treatment. *Scand J Rehabil Med.* 1989;21:91-95.
48. Menke JM. Do manual therapies help low back pain? A comparative effectiveness meta-analysis. *Spine (Phila Pa 1976).* 2014;39:E463-E472. <https://doi.org/10.1097/BRS.0000000000000230>
49. Mieritz RM, Bronfort G, Hartvigsen J. Regional lumbar motion and patient-rated outcomes: a secondary analysis of data from a randomized clinical trial. *J Manipulative Physiol Ther.* 2014;37:628-640. <https://doi.org/10.1016/j.jmpt.2014.10.004>
50. Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med.* 2009;6:e1000097. <https://doi.org/10.1371/journal.pmed.1000097>
51. Moola S, Munn Z, Tufanaru C, et al. Systematic

reviews of etiology and risk. In: Aromataris E, Munn Z, eds. *Joanna Briggs Institute Reviewer's Manual*. Adelaide, Australia: Joanna Briggs Institute; 2017:ch 7.

52. Mueller S, Stoll J, Mueller J, Cassel M, Mayer F. Trunk muscle activity during drop jump performance in adolescent athletes with back pain. *Front Physiol*. 2017;8:274. <https://doi.org/10.3389/fphys.2017.00274>
53. Munn Z, Moola S, Riitano D, Lisy K. The development of a critical appraisal tool for use in systematic reviews addressing questions of prevalence. *Int J Health Policy Manag*. 2014;3:123-128. <https://doi.org/10.15171/ijhpm.2014.71>
54. Nordstoga AL, Meisingset I, Vasseljen O, Nilsen TIL, Unsgaard-Tøndel M. Longitudinal associations of kinematics and fear-avoidance beliefs with disability, work ability and pain intensity in persons with low back pain. *Musculoskelet Sci Pract*. 2019;41:49-54. <https://doi.org/10.1016/j.msksp.2019.03.008>
55. Olugbade T, Bianchi-Berthouze N, Williams ACC. The relationship between guarding, pain, and emotion. *Pain Rep*. 2019;4:e770. <https://doi.org/10.1097/PR9.0000000000000770>
56. O'Sullivan K, O'Sullivan P, Vibe Fersum K, Kent P. Better targeting care for individuals with low back pain: opportunities and obstacles. *Br J Sports Med*. 2017;51:489-490. <https://doi.org/10.1136/bjsports-2016-096612>
57. O'Sullivan P. Diagnosis and classification of chronic low back pain disorders: maladaptive movement and motor control impairments as underlying mechanism. *Man Ther*. 2005;10:242-255. <https://doi.org/10.1016/j.math.2005.07.001>
58. Osumi M, Sumitani M, Otake Y, et al. Kinesiophobia modulates lumbar movements in people with chronic low back pain: a kinematic analysis of lumbar bending and returning movement. *Eur Spine J*. 2019;28:1572-1578. <https://doi.org/10.1007/s00586-019-06010-4>
59. Pagé I, Marchand AA, Nougareux F, O'Shaughnessy J, Descarreaux M. Neuromechanical responses after biofeedback training in participants with chronic low back pain: an experimental cohort study. *J Manipulative Physiol Ther*. 2015;38:449-457. <https://doi.org/10.1016/j.jmpt.2015.08.005>
60. Poitras S, Loisel P, Prince F, Lemaire J. Disability measurement in persons with back pain: a validity study of spinal range of motion and velocity. *Arch Phys Med Rehabil*. 2000;81:1394-1400. <https://doi.org/10.1053/apmr.2000.9165>
61. Qiao J, Zhang SL, Zhang J, Feng D. A study on the paraspinal muscle surface electromyography in acute nonspecific lower back pain. *Medicine (Baltimore)*. 2019;98:e16904. <https://doi.org/10.1097/MD.00000000000016904>
62. Shahvarpour A, Henry SM, Preuss R, Mecheri H, Larivière C. The effect of an 8-week stabilization exercise program on the lumbopelvic rhythm and flexion-relaxation phenomenon. *Clin Biomech (Bristol, Avon)*. 2017;48:1-8. <https://doi.org/10.1016/j.clinbiomech.2017.06.010>
63. Siemieniuk R, Guyatt G. What is GRADE? Available at: <https://bestpractice.bmj.com/info/us/toolkit/learn-ebm/what-is-grade/>. Accessed May 22, 2019.
64. Steiger F, Wirth B, de Bruin ED, Mannion AF. Is a positive clinical outcome after exercise therapy for chronic non-specific low back pain contingent upon a corresponding improvement in the targeted aspect(s) of performance? A systematic review. *Eur Spine J*. 2012;21:575-598. <https://doi.org/10.1007/s00586-011-2045-6>
65. Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ*. 2019;366:l4898. <https://doi.org/10.1136/bmj.l4898>
66. Strand LI, Anderson B, Lygren H, Skouen JS, Ostelo R, Magnussen LH. Responsiveness to change of 10 physical tests used for patients with back pain. *Phys Ther*. 2011;91:404-415. <https://doi.org/10.2522/ptj.20100016>
67. van der Hulst M, Vollenbroek-Hutten MM, Rietman JS, Schaake L, Groothuis-Oudshoorn KG, Hermens HJ. Back muscle activation patterns in chronic low back pain

during walking: a "guarding" hypothesis. *Clin J Pain*. 2010;26:30-37. <https://doi.org/10.1097/AJP.0b013e3181b40eca>

68. Vibe Fersum K, O'Sullivan P, Skouen JS, Smith A, Kvåle A. Efficacy of classification-based cognitive functional therapy in patients with non-specific chronic low back pain: a randomized controlled trial. *Eur J Pain*. 2013;17:916-928. <https://doi.org/10.1002/j.1532-2149.2012.00252.x>
69. Watson PJ, Booker CK, Main CJ. Evidence for the role of psychological factors in abnormal paraspinal activity in patients with chronic low back pain. *J Musculoskelet Pain*. 1997;5:41-56. https://doi.org/10.1300/J094v05n04_05
70. Watson PJ, Booker CK, Main CJ, Chen AC. Surface electromyography in the identification of chronic low back pain patients: the development of the flexion relaxation ratio. *Clin Biomech (Bristol, Avon)*. 1997;12:165-171. [https://doi.org/10.1016/s0268-0033\(97\)00065-x](https://doi.org/10.1016/s0268-0033(97)00065-x)
71. Webb R, Brammah T, Lunt M, Urwin M, Allison T, Symmons D. Prevalence and predictors of intense, chronic, and disabling neck and back pain in the UK general population. *Spine (Phila Pa 1976)*. 2003;28:1195-1202. <https://doi.org/10.1097/01.BRS.0000067430.49169.01>
72. Williams CM, Kamper SJ. Non-specific effects of acupuncture – does the 'placebo' effect play an important role? *Br J Sports Med*. 2012;46:578-579. <https://doi.org/10.1136/bjsports-2012-091229>
73. Xia T, Long CR, Vining RD, et al. Association of lumbar spine stiffness and flexion-relaxation phenomenon with patient-reported outcomes in adults with chronic low back pain – a single-arm clinical trial investigating the effects of thrust spinal manipulation. *BMC Complement Altern Med*. 2017;17:303. <https://doi.org/10.1186/s12906-017-1821-1>



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SEARCH STRATEGY USED IN THE MEDLINE DATABASE

1. Case Reports/
2. (case report* or case series or case-series or single-case report).ti,ab,pt.
3. 1 or 2
4. Low Back Pain/
5. (Low* back pain or low* back ache or low* backache or lumbar pain or lumbar spin* pain or backache or lumbago or low* back dysfunction or spondylosis or dorsalgia or back strain or back pain or spin* pain or low* back syndrome or lumbopelvic pain or lumbo-pelvic pain or NSLBP or CLBP or NSCLBP or Persistent low* back pain or PLBP).ti,ab.
6. 4 or 5
7. exp Movement/ or exp Biomechanical Phenomena/ or Range of Motion, Articular/
8. (Kinematic* or Movement* or movement pattern* or ROM or Range of motion* or Range of movement* or excursion or movement control or motor control or stabilisation or stability or angle or lordosis angle or hip contribution or lumbar contribution or pelvi* contribution or position or relative position or joint position or dynamic position or dynamic posture or speed or velocity or timing or flexibility or mobility or muscle activ* or magnitude or EMG or sEMG or electromyography or electromyogram or surface electromyogra* or muscle activation patterns).ti,ab.
9. (flexion relaxation or flexion-relaxation).mp.
10. ((flexion relaxation or flexion-relaxation or FR) and (ratio or response or phenomena or phenomenon)).ti,ab.
11. or/7-10
12. 3 and 6 and 11
13. exp animals/ not humans.sh.
14. 12 not 13
15. reliability.ti. or cross-sectional.pt. or cross sectional.pt. or pregnancy.ti,ab.
16. spinal neoplasms/
17. (cauda equin* or cancer or metastasis or metabolic disorder or neuropathy or surg* or fracture*).mp.
18. 15 or 16 or 17
19. 14 not 18
20. limit 19 to English language
21. Case-control studies/ or cohort studies/ or comparative study/ or evaluation studies/ or follow-up studies/ or longitudinal studies/ or prospective studies/ or treatment outcome/ or exp Clinical Study/ or Case Reports/ or observational study/
22. (cohort or compared or comparison or groups or Case-control or case control or follow up or follow-up studies or treatment outcome or case reports or case series or case-series or case-cohort or case cohort).ti,ab,pt.
23. Randomised Controlled Trials/ or randomi?ed controlled trial*.pt,ti. or randomi?ed controlled study.ti. or randomi?ed clinical trial.ti. or randomi?ed clinical study.ti. or conference proceedings.pt,ti,ab.
24. or/21-23
25. Low Back Pain/
26. (Low* back pain or low* back ache or low* backache or lumbar pain or lumbar spin* pain or backache or lumbago or low* back dysfunction or spondylosis or dorsalgia or back strain or back pain or spin* pain or low* back syndrome or lumbopelvic pain or lumbo-pelvic pain or NSLBP or CLBP or NSCLBP or Persistent low* back pain or PLBP).ti,ab.
27. 25 or 26
28. exp Movement/ or exp Biomechanical Phenomena/ or Range of Motion, Articular/
29. (Kinematic* or Movement* or movement pattern* or ROM or Range of motion* or Range of movement* or excursion or movement control or motor control or stabilisation or stability or angle or lordosis angle or hip contribution or lumbar contribution or pelvi* contribution or position or relative position or joint position or dynamic position or dynamic posture or speed or velocity or timing or flexibility or mobility or muscle activ* or magnitude or EMG or sEMG or electromyography or electromyogram or surface electromyogra* or muscle activation patterns).ti,ab.
30. (flexion relaxation or flexion-relaxation).mp.
31. ((flexion relaxation or flexion-relaxation or FR) and (ratio or response or phenomena or phenomenon)).ti,ab.
32. 30 or 31
33. 28 or 29 or 32
34. (correlat* or regression or association or covarian*).ti,ab.
35. ((chang* or improv* or modif* or alter* or increase or decrease or reduction or outcome) adj10 (correlat* or associat* or relat* or contingent)).ti,ab.
36. (explain adj3 varian*).ti,ab.
37. 34 or 35 or 36
38. 24 and 27 and 33 and 37

APPENDIX A

39. exp animals/ not humans.sh.
40. 38 not 39
41. reliability.ti. or cross-sectional.pt. or cross sectional.pt. or pregnancy.ti.ab.
42. spinal neoplasms/
43. (cauda equin* or cancer or metastasis or metabolic disorder or neuropathy or surg* or fracture*).mp.
44. 41 or 42 or 43
45. 40 not 44
46. limit 45 to English language
47. 20 or 46

APPENDIX B

ADAPTED JOANNA BRIGGS INSTITUTE CRITICAL APPRAISAL TOOL⁵¹ FOR COHORT STUDIES^a

Risk-of-Bias Item	Description
1. Were the demographic and clinical characteristics of the participants adequately described?	Description of the participants adequate enough to understand how generalizable (externally valid) the findings might be to other settings
2. Were the selection criteria for participants adequately described?	Knowledge of how the participants were selected is important to also understand how generalizable the findings are (eg, the sample only included participants with certain traits, such as generalized hypermobility)
3. Was there an acceptable missingness of data?	Data missingness is not in excess of 20% (greater than 80% of the data were analyzed). When applicable, statistical analysis that handles missing data well is used (eg, generalized estimating equations, mixed-effect models, latent growth models). Nonrandom missingness from people in a cohort can reduce external validity, while missingness of data time points in longitudinal studies can reduce internal validity
4. Were the movements assessed in a standardized and valid way?	The study clearly describes the method of measurement of movements. Assessing validity requires that a "pseudo-gold standard" is available and to which the measure has been compared, or that construct and concurrent validity have been quantified in some other way. The measurement needs to be well described and referenced unless it is well known
5. Were movements assessed in a reliable way?	In this context, reliability (reproducibility) refers to the quantification of test-retest variability. The measure needs to be well known or, if it is not well known, referenced
6. Were pain and/or activity limitation assessed in a standardized and valid way?	The study should clearly describe the method of measurement of pain and/or activity limitation. Validity requires some demonstration (eg, a reference) of construct and concurrent validity
7. Were pain and/or activity limitation assessed in a reliable way?	Reliability refers here to the quantification of test-retest variability. The method of measurement needs to be well known or, if it is not well known, referenced
8. Was appropriate statistical analysis used?	Transparency and appropriate selection of the analytical strategy used. In this case, appropriate statistical approaches would include forms of correlation, regression, and visual analysis
9. Was the assessment of movement (and its change) blind to the assessment of pain or activity limitation (and its change), or vice versa?	Assessor knowledge of participant self-report of pain or activity limitation (or its change) may have biased the assessor's subsequent assessment of movement (or vice versa). If assessor blinding is reported, a score of 1 ("yes") is given; if nonblinding of assessors is reported, a score of 0 ("no") is given; and if it is not clear whether blinding of assessors took place, a rating of "unclear" is given

^aEach item is scored "yes," "no," "unclear," or "not applicable."

SCORES AND REASONING FOR THE 5 GRADE DOMAINS ASSESSING QUALITY OF EVIDENCE ACROSS STUDIES

The GRADE guidelines suggest starting at “medium quality” for research using cohort designs, and we used the other GRADE criteria to upgrade or downgrade the quality score.

Domain 1: Risk of Bias

Of the individual studies included in this review, 93% were categorized as low quality for our research question, and only 7% as high quality. For many studies, this categorization was due to uncertainty about the blinding of assessors, because this aspect was unreported.

Domain 2: Imprecision

The inclusion of over 2700 individuals with nonspecific low back pain from 27 different studies improves the generalizability and precision of the systematic review. However, the use of movement measurement tools that have uncertain within- and between-session error resulted in a lack of confidence in identifying true change.

Domain 3: Inconsistency

Infrequently identifying a relationship among both pain and activity limitation outcomes, relative consistency among the significant r values detailed in **APPENDIX D**, and the very consistent findings regarding the direction of movement change in the presence of a relationship (92.5%) increase our confidence in the findings.

Domain 4: Indirectness

Variable study designs and variable intervention types and durations, as well as multiple different outcome measures (although generalizable to what would be expected in clinical practice), resulted in heterogeneous studies and populations. Given the likelihood that clinically relevant movements vary across this population and the frequent use of study designs and interventions that were not readily able to identify and target person-specific salient movements, we believe that this often resulted in an inability to investigate the relationship directly.

Domain 5: Publication Bias

Selective publication of studies that show a relationship between changes in movement and pain or activity limitation would be likely to overestimate the presence of a relationship. However, we included studies that did not directly target movement, which may minimize the influence of overestimation.

Criteria That Could Influence a Decision to Upgrade the Quality Rating

1. *Large Effect*
Not applicable to this systematic review.
2. *Dose Response*
Not applicable to this systematic review.
3. *Accounted for All Plausible Residual Confounding*
Not applicable to this systematic review.

Summary

Starting with a GRADE rating of “medium quality” for cohort-derived data, after consideration of all GRADE criteria, we downgraded the classification to low-quality evidence, defined by GRADE as “our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.”

The main reason was because, for our research question about a relationship of change between clinical attributes, our judgment is that the degree of indirectness in the included studies and the possibility of a lack of movement assessor blinding resulted in a meaningful threat to the estimate of how frequently that relationship occurs.

Abbreviation: GRADE, Grading of Recommendations Assessment, Development and Evaluation.

References

- Balslem H, Helfand M, Schünemann HJ, et al. GRADE guidelines: 3. Rating the quality of evidence. *J Clin Epidemiol*. 2011;64:401-406. <https://doi.org/10.1016/j.jclinepi.2010.07.015>
- Guyatt GH, Oxman AD, Vist GE, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. *BMJ*. 2008;336:924-926. <https://doi.org/10.1136/bmj.39489.470347.AD>
- Siemieniuk R, Guyatt G. What is GRADE? Available at: <https://bestpractice.bmj.com/info/us/toolkit/learn-ebm/what-is-grade/>. Accessed May 22, 2019.

APPENDIX E

BASELINE PAIN AND ACTIVITY LIMITATION LEVELS FOR STUDIES THAT INVESTIGATED RELATIONSHIPS BETWEEN CHANGES IN FLEXION-RELATED MOVEMENT PARAMETERS AND CHANGES IN PAIN OR ACTIVITY LIMITATION AT THE INDIVIDUAL LEVEL

Data Source	Design/Outcome Variable	Relationship	No Relationship
Baseline values for studies included in this systematic review	Pain (0-100) ^a	54 (54-54)	49 (21-74)
	Studies, n	1	8
	IQR ^b	0	7
	Participants per study ^a	148	33 (17-68)
	Activity limitation (0-100) ^a	32 (24-37)	30 (13-49)
	Studies, n	10	6
	IQR ^b	8	11
	Participants per study ^a	54.5 (18-148)	78 (21-111)
Baseline values for trials in individuals with NSLBP ^c	Baseline pain (0-100)	48 ± 14 (51)	
	Baseline activity limitation (0-100)	41 ± 12 (46)	

Abbreviations: IQR, interquartile range; NSLBP, nonspecific low back pain.

^aValues are median or median (range).

^bThe IQR was calculated as the upper quartile minus the lower quartile.

^cValues are mean ± SD (median). The following studies were used for comparison:

Brinkhaus B, Witt CM, Jena S, et al. Acupuncture in patients with chronic low back pain: a randomized controlled trial. Arch Intern Med. 2006;166:450-457. <https://doi.org/10.1001/archinte.166.4.450>

Fairbank J, Frost H, Wilson-MacDonald J, et al. Randomised controlled trial to compare surgical stabilisation of the lumbar spine with an intensive rehabilitation programme for patients with chronic low back pain: the MRC spine stabilisation trial. BMJ. 2005;330:1233. <https://doi.org/10.1136/bmj.38441.620417.8F>

Frost H, Moffett JAK, Moser JS, Fairbank JC. Randomised controlled trial for evaluation of fitness programme for patients with chronic low back pain. BMJ. 1995;310:151-154. <https://doi.org/10.1136/bmj.310.6973.151>

Hill JC, Whitehurst DG, Lewis M, et al. Comparison of stratified primary care management for low back pain with current best practice (STarT Back): a randomised controlled trial. Lancet. 2011;378:1560-1571. [https://doi.org/10.1016/S0140-6736\(11\)60937-9](https://doi.org/10.1016/S0140-6736(11)60937-9)

Kamper SJ, Apeldoorn AT, Chiarotto A, et al. Multidisciplinary biopsychosocial rehabilitation for chronic low back pain. Cochrane Database Syst Rev. 2014;CD000963. <https://doi.org/10.1002/14651858.CD000963.pub3>

Moffett JK, Torgerson D, Bell-Syer S, et al. Randomised controlled trial of exercise for low back pain: clinical outcomes, costs, and preferences. BMJ. 1999;319:279-283. <https://doi.org/10.1136/bmj.319.7205.279>

UK BEAM Trial Team. United Kingdom back pain exercise and manipulation (UK BEAM) randomised trial: effectiveness of physical treatments for back pain in primary care. BMJ. 2004;329:1377. <https://doi.org/10.1136/bmj.38282.669225.AE>

Vibe Fersum K, O'Sullivan P, Skouen JS, Smith A, Kvåle A. Efficacy of classification-based cognitive functional therapy in patients with non-specific chronic low back pain: a randomized controlled trial. Eur J Pain. 2013;17:916-928. <https://doi.org/10.1002/j.1532-2149.2012.00252.x>

[LITERATURE REVIEW]

APPENDIX D

RESULTS FROM INDIVIDUAL STUDIES

Study, n, Measurement Type, Measurement Tool	Length of LBP, Time Between Measures	Movement Reported for Cohort Studies ^a		Pain Reported for Cohort Studies ^a			Activity Limitation Reported for Cohort Studies ^a		
		Pre	Post	Pre	Post	Related?	Pre	Post	Related?
		Flexion ROM (peak)							
Beladev and Masharawi ⁶ n = 25 Lumbar Dual inclinometer	>12 wk 8 wk	22.45° ± 8.27° (range, 7.15°-35.8°)	28.55° ± 5.20° (range, 19°-35.30°)	VAS: 7.38 ± 1.62 (range, 5-10)	VAS: 3.17 ± 2.06 (range, 0-7)	No: “change in VAS score was not correlated with changes in lumbar flexion”	...	...	...
Helmhout et al ¹⁹ n = 90 Lumbar Spinal mouse inclinometer	>2 y 11 wk	NA	NA	NA	NA	No: “improvements in outcome did not relate to improvement in lumbar mobility”	NA	NA	No: “improvements in outcome did not relate to improvements in lumbar mobility”
Pagé et al ¹⁹ n = 21 Lumbar Motion-tracking system	88.36 ± 65.36 mo (range, 12-240 mo) 4-6 wk	NA	NA	NA	NA	No (<i>r</i> = 0.04, <i>P</i> = .703) for a VAS with flexion ROM	NA	NA	No: changes in flexion did not contribute significantly to the models that explained change in disability QBPDs: <i>r</i> = 0.02, <i>P</i> = .830 PSFS: <i>r</i> = 0.11, <i>P</i> = .318
Shahvarpour et al ¹² n = 33 Both (lumbar, pelvic, and thoracic ROM) Motion sensors	>4 wk 8 wk	NA	NA	NA	NA	No Lumbar: <i>r</i> = −0.04, <i>P</i> = .826 Thoracic: <i>r</i> = 0.01, <i>P</i> = .954 Pelvis: <i>r</i> NA, <i>P</i> ≥ .1	NA	NA	1/3 showed a relationship Lumbar: <i>r</i> = 0.42, <i>P</i> = .015 Thoracic: <i>r</i> = 0.19, <i>P</i> = .297 Pelvis: <i>r</i> NA, <i>P</i> ≥ .1
Caporaso et al ⁹ n = 32 Multisegment Fingertip to floor	8.6 ± 11.1 y 9 wk	NA	NA	...	...	...	NA	NA	Yes RMDQ: <i>r</i> = 0.63, <i>P</i> < .001
Ekedahl et al ¹¹ n = 65 Multisegment Fingertip to floor	>12 wk 4 wk	24 ± 16 cm	Change: 12 ± 3 cm	...	...	...	RMDQ: 11.2 ± 5.6	Change: 5.2 ± 5.4	Yes RMDQ: <i>r</i> = 0.66, <i>P</i> < .001
Hurri et al ²³ n = 88 Lumbar 2-goniometer method	9.7 ± 7.3 y 3 wk	43° ± 10°	48° ± 10°	...	...	...	RMDQ: 29 ± 15	RMDQ: 20 ± 16	No (<i>r</i> = −0.02, NS)

Table continues on page A2.

[LITERATURE REVIEW]

APPENDIX D

Study, n, Measurement Type, Measurement Tool	Length of LBP, Time Between Measures	Movement Reported for Cohort Studies ^a		Pain Reported for Cohort Studies ^a			Activity Limitation Reported for Cohort Studies ^a		
		Pre	Post	Pre	Post	Related?	Pre	Post	Related?
Mannion et al ⁴⁰ n = 148 Lumbar Electromechanical device	... 3 mo	NA	NA	NA	NA	Yes ($r = 0.30, P = .0006$)	NA	NA	Yes ($r = 0.18, P = .04$)
Mellin et al ⁴⁷ n = 456 Lumbar Dual inclinometer	NA 3 wk	NA	NA	...	...	...	NA	NA	No Male, $r = 0.05, NS$; female, $r = 0.07, NS$
Strand et al ⁶⁶ n = 98 Both Kyphometer and fingertip to floor	"Long lasting" 3-5 wk	Spondylometry: $96.6^\circ \pm 18.7^\circ$ Fingertip to floor: 13.8 ± 15.5 cm	Spondylometry: $103.1^\circ \pm 17.8^\circ$ Fingertip to floor: 8.4 ± 13.4 cm	...	...	...	HFQ: 10.2 ± 4.4 RMDQ: 11.8 ± 4.4	HFQ: 7.9 ± 5.3 RMDQ: 8.9 ± 5.5	3/4 showed a relationship HFQ and spondylometry: $r = 0.27, P < .05$ HFQ and fingertip to floor: $r = 0.44, P < .01$ RMDQ and fingertip to floor: $r = 0.22, P < .05$ RMDQ and spondylometry: $r = 0.08, P > .05$
Mannion et al ³⁸ n = 32 Multisegment Fingertip to floor	>3 mo 9 wk	NA	NA	VAS: 46 (estimated from graph)	VAS: 36 (estimated from graph)	NA	RMDQ: 9.12 (estimated from graph)	RMDQ: 6.48 (estimated from graph)	Yes ($r = 0.45, P = .009$)
Nordstoga et al ⁶⁴ n = 44 Multisegment Motion sensor	Any length (62% had pain for >9 mo) 3 mo	ROM: $119^\circ \pm 25.1^\circ$ Peak flexion velocity: start, $61.0^\circ/s \pm 22.6^\circ/s$; middle, $62.8^\circ/s \pm 25.4^\circ/s$; end, $47.2^\circ/s \pm 22.6^\circ/s$	Change in ROM: 10.3° (95% CI: $1.4^\circ, 19.2^\circ$) Peak flexion velocity: start, $18.7^\circ/s$ (95% CI: $10.4^\circ/s, 27.1^\circ/s$); middle, $17.9^\circ/s$ (95% CI: $9.4^\circ/s, 26.4^\circ/s$); end, $16.1^\circ/s$ (95% CI: $7.8^\circ/s, 24.5^\circ/s$)	VAS: 5.4 ± 2.2	VAS: 2.9 ± 2.1	No: changes in ROM or peak velocity were not associated with changes in pain	ODI: 23.6 ± 12.4	ODI: 14.4 ± 12.1	Yes: 1° increase in ROM was weakly associated with less activity limitation (-0.14 points; 95% CI: $-0.22, 0.06$) Changes in peak velocity were not associated with changes in activity limitation
Poitras et al ⁶⁰ n = 111 Multisegment Optoelectric system	Subacute 28 wk	NA	NA	...	...	...	ODI: 30.4 ± 15.5	ODI: 15.2 ± 14.3	Reported no relationship; no correlation values given

Table continues on page A3.

APPENDIX D

Study, n, Measurement Type, Measurement Tool	Length of LBP, Time Between Measures	Movement Reported for Cohort Studies*		Pain Reported for Cohort Studies*			Activity Limitation Reported for Cohort Studies*		
		Pre	Post	Pre	Post	Related?	Pre	Post	Related?
Demoulin et al ¹⁰ n = 17 Both Rachimetry and modified Schober test	11 ± 9 y 6 wk	Spinal flexion (rachimetry): 40.2 ± 8.4 Schober test: 6.7 ± 1.4	Spinal flexion (rachimetry): 40.8 ± 12.8 Schober test: 7.1 ± 1.4	VAS: 43.6 ± 23	VAS: 18.9 ± 14	No	...	...	...
<i>Flexion EMG</i>									
Marshall and Murphy ⁴¹ n = 18 Lumbar sEMG: L4-5 (FRR)	4.8 y 12 wk	3.26 ± 3.43	6.53 ± 3.34	VAS: 47.39 ± 23.72	VAS: 23.94 ± 19.71	Did not assess	ODI: 24% (estimated from graph)	ODI: 10% (estimated from graph)	Yes: "increases in the FRR best explained changes measured by the ODI.... The change in erector spinae FRR explained 38% of the variance in the change in ODI score"
Pagé et al ⁶⁹ n = 21 Lumbar sEMG: L3 bilaterally (FRR)	88.36 ± 65.36 mo (range, 12-240 mo) 4-6 wk	NA	NA	NA	NA	No significant relationship was found between changes in clinical outcome and neuro- mechanical variable	NA	NA	No significant relation- ship was found between changes in clinical outcome and neuromechanical variable
Qiao et al ⁶¹ n = 45 Lumbar sEMG: 2 cm lateral to L4	Acute NSLBP 1 wk	Flexion: 68.79 ± 19.56 μ V Maintained flexion: 72.38 ± 12.39 μ V Extension: 87.82 ± 21.48 μ V Upright: 20.98 ± 7.51 μ V	Flexion: 60.28 ± 17.26 μ V Maintained flexion: 28.79 ± 11.1 μ V Extension: 110.65 ± 23.16 μ V Upright: 20.10 ± 7.13 μ V	VAS: 5.15 ± 1.58	VAS: 2.05 ± 1.03	1/4 reported a significant correlation Flexion: r = 0.187, P = .312 Maintained flexion: r = 0.869, P ≤ .001 Extension: r = 0.294, P = .114 Upright: NS	...	...	...
Shahvarpour et al ⁶² n = 33 Lumbar sEMG: longissimus, iliocostalis, and multifidus (FRR)	>4 wk 8 wk	NA	NA	NA	NA	No (r = -0.31, P = .066)	NA	NA	No (r = -0.18, P = .283)

Table continues on page A4.

[LITERATURE REVIEW]

APPENDIX D

Study, n, Measurement Type, Measurement Tool	Length of LBP, Time Between Measures	Movement Reported for Cohort Studies ^a		Pain Reported for Cohort Studies ^a		Related?	Activity Limitation Reported for Cohort Studies ^a		
		Pre	Post	Pre	Post		Pre	Post	Related?
Watson et al ⁶⁹ n = 36 Lumbar sEMG: L1-2, L4-5 bilaterally (FRR)	4.6 ± 4.2 y 3 wk	Left L1-2, 2.9 ± 4.5; right L1-2, 2.6 ± 2.9 Left L4-5, 2.3 ± 3.3; right L4-5, 2.6 ± 4.4	Left L1-2, 6.5 ± 10.4; right L1-2, 5.4 ± 7.7 Left L4-5, 4.1 ± 5.8; right L4-5, 3.7 ± 4.8	VAS: 49.6 ± 22.6	NA	No Left L1-2: $r = -0.10$, NS Right L1-2: $r = -0.12$, NS Left L4-5: $r = -0.15$, NS Right L4-5: $r = -0.15$, NS	ODI: 51.1 ± 13.9	NA	NA
Xia et al ⁷³ n = 68 Lumbar sEMG: paraspinals (FRR)	>12 wk (84% had LBP for >1 y) 6 wk	4.7 ± 3.7	5.5 ± 5.6	VAS: 46.1 ± 18.1	VAS: 25.6 ± 25.4	No: "the addition of each physiologic variable did not affect the parameter estimates for LBP intensity over time"	RMDQ: 9.5 ± 4.3	RMDQ: 4.8 ± 4.0	No: "the addition of each physiologic variable did not affect the parameter estimates for LBP-related disability over time"
<i>Flexion velocity/lumbopelvic rhythm</i>									
Shahvarpour et al ⁶² n = 33 Lumbar Motion sensors (% lumbar contribution)	>4 wk 8 wk	NA	NA	NA	NA	No ($r = -0.02$, $P = .908$)	NA	NA	Yes ($r = 0.4$, $P = .019$)
Nordstoga et al ⁶⁴ n = 44 Multisegment Motion sensor (velocity)	Any length (62% had pain for >9 mo) 3 mo	Peak flexion velocity: start, 61.0°/s ± 22.6°/s; middle, 62.8°/s ± 25.4°/s; end, 47.2°/s ± 22.6°/s	Change in peak flexion velocity: start, 18.7°/s (95% CI: 10.4°/s, 27.1°/s); middle, 17.9°/s (95% CI: 9.4°/s, 26.4°/s); end, 16.1°/s (95% CI: 7.8°/s, 24.5°/s)	VAS: 5.4 ± 2.2	VAS: 2.9 ± 2.1	No: changes in ROM or peak velocity were not associated with changes in pain	ODI: 23.6 ± 12.4	ODI: 14.4 ± 12.1	No: changes in peak velocity were not associated with changes in activity limitation

Table continues on page A5.

APPENDIX D

Study, n, Measurement Type, Measurement Tool	Length of LBP, Time Between Measures	Movement Reported for Cohort Studies ^a		Pain Reported for Cohort Studies ^a			Activity Limitation Reported for Cohort Studies ^a		
		Pre	Post	Pre	Post	Related?	Pre	Post	Related?
Poitras et al ⁶⁰ n = 111 Multisegment Optoelectric system (7 velocity/ rhythm param- eters)	Subacute 28 wk	NA	NA	...	...	...	NA	NA	Yes for maximum pelvis velocity ($r = 0.51, P = .002$) All others were NS and not reported
Shahvarpour et al ⁶² n = 33 Multisegment Motion sensors (% pelvis contribu- tion)	>4 wk 8 wk	NA	NA	NA	NA	No ($r = -0.04, P = .822$)	NA	NA	No ($r = -0.31, P = .081$)
<i>Extension ROM (peak)</i>									
Beladev and Masharawi ⁶ n = 25 Lumbar Dual inclinometer	>12 wk 8 wk	763° ± 4.45° (range, 2.15°- 17.15°)	11.14° ± 3.32° (range, 5.3°-20°)	VAS: 7.38 ± 1.62 (range, 5-10)	VAS: 3.17 ± 2.06 (range, 0-7)	No: "change in VAS score was not cor- related with changes in lumbar extension"	...	...	...
Helmhout et al ¹⁹ n = 90 Lumbar Spinal mouse inclinometer	>2 y 11 wk	NA	NA	NA	NA	No ($r = -0.06, P = .624$)	NA	NA	No QBPDs: $r = -0.02, P = .855$ PSC: $r = -0.14, P = .211$
Hurri et al ²³ n = 88 Lumbar 2-goniometer method	9.7 ± 7.3 y 3 wk	17° ± 9°	16° ± 7°	...	...	...	RMDQ: 29 ± 15	RMDQ: 20 ± 16	No ($r = 0.02, NS$)
Mellin et al ⁴⁷ n = 456 Lumbar Dual inclinometer	NA 3 wk	NA	NA	...	...	...	NA	NA	No Male: $r = -0.02, NS$; female: $r = 0.10, NS$
<i>Lateral flexion ROM (peak)</i>									
Elnaggar et al ¹² n = 56 Lumbar Motion-tracking system	>3 mo 2 wk	Flexion exer- cise group: 98 ± 19.4 Extension exercise group: 97 ± 18.3	Flexion exercise group: 99.4 ± 16.5 Extension exercise group: 98.6 ± 21.1	Flexion exer- cise group: modified MPQ, 14.1 ± 9.8 Extension exercise group: modified MPQ, 15.9 ± 7.8	Flexion exer- cise group: modified MPQ, 8.9 ± 9.4 Extension exercise group: modified MPQ, 10.6 ± 8.6	Yes ($r = 0.35, P = .008$)	...	...	...

Table continues on page A6.

[LITERATURE REVIEW]

APPENDIX D

Study, n, Measurement Type, Measurement Tool	Length of LBP, Time Between Measures	Movement Reported for Cohort Studies ^a		Pain Reported for Cohort Studies ^a			Activity Limitation Reported for Cohort Studies ^a		
		Pre	Post	Pre	Post	Related?	Pre	Post	Related?
Mellin et al ⁴⁷ n = 456 Lumbar Dual inclinometer	>2 y 3 wk	NA	NA	...	...	...	NA	NA	Yes Male: $r = 0.12$, $P < .05$; female: $r = 0.16$, $P < .05$
Anderson et al ² n = 96 Multisegment Fingertip to floor	>3 mo 3.5 wk	14.0 ± 4.5 cm	Change: 2.8 ± 3.6 cm	NRS: 5.9 ± 1.8	NRS change: 1.2 ± 1.8	Yes ($r = 0.272$): "weak correlation" reported. No mention of P value	RMDQ: 11.8 ± 4.4	RMDQ change: 2.8 ± 4.2	No ($r = 0.194$): "no correlation" reported. No mention of P value
Hurri et al ²³ n = 88 Multisegment Fingertip distance change	9.7 ± 7.3 y 3 wk	175 ± 38 mm	190 ± 45 mm	...	...	...	RMDQ: 29 ± 15	RMDQ: 20 ± 16	Yes ($r = -0.30$, $P < .01$)
Strand et al ⁶⁶ n = 98 Multisegment Fingertip to floor	"Long lasting" 3-5 wk	Fingertip to floor, 14.0 ± 4.6 cm	Fingertip to floor, 16.8 ± 5.0 cm Change: 2.8 ± 3.7 cm (95% CI: 2.1, 3.5 cm)	...	...	...	HFQ: 10.2 ± 4.4 RMDQ: 11.8 ± 4.4	HFQ: 7.9 ± 5.3 RMDQ: 8.9 ± 5.5	1/2 showed a relationship HFQ: $r = 0.22$, $P < .05$ RMDQ: $r = 0.18$, $P > .05$
Rotation ROM (peak)									
Elnaggar et al ¹² n = 56 Lumbar Motion-tracking system	>3 mo 2 wk	Flexion exercise group: 88.6 ± 18.4 Extension exercise group: 86 ± 18.2	Flexion exercise group: 92.4 ± 18.8 Extension exercise group: 90 ± 20.3	Flexion exercise group: modified MPQ, 14.1 ± 9.8 Extension exercise group: modified MPQ, 15.9 ± 7.8	Flexion exercise group: modified MPQ, 8.9 ± 9.4 Extension exercise group: modified MPQ, 10.6 ± 8.6	Yes ($r = 0.35$, $P = .008$)	...	...	...
Hurri et al ²³ n = 88 Lumbar 2-goniometer method	9.7 ± 7.3 y 3 wk	45° ± 11°	49° ± 11°	...	...	...	RMDQ: 29 ± 15	RMDQ: 20 ± 16	Yes ($r = -0.24$, $P < .05$)
Mellin et al ⁴⁷ n = 456 Multisegment Compass	>2 y 3 wk	NA	NA	...	...	...	NA	NA	Yes Male: $r = 0.12$, $P < .05$; female: $r = 0.31$, $P < .01$

Table continues on page A7

APPENDIX D

Study, n, Measurement Type, Measurement Tool	Length of LBP, Time Between Measures	Movement Reported for Cohort Studies ^a		Pain Reported for Cohort Studies ^a			Activity Limitation Reported for Cohort Studies ^a		
		Pre	Post	Pre	Post	Related?	Pre	Post	Related?
Summative ROM (peak)									
Mieritz et al ⁴⁹ n = 199 Lumbar Spinal motion analyzer (6-DOF linkage system)	>6 wk (82% for >1 y) 12 wk	NA	NA	NA	NA	2-D circum- duction: r = -0.22, $P < .01$ 3-D circum- duction: r = -0.09, $P > .01$ Sagittal ROM: r = -0.09, $P > .01$	NA	NA	2-D circumduction: r = -0.20, $P < .01$ 3-D circumduction: r = -0.14, $P > .01$ Sagittal ROM: r = -0.15, $P > .01$
Martin et al ⁴² n = 36 Lumbar Vector stereograph	>6 wk 4 wk	NA	NA	NA	NA	No: correla- tions were 0.00 and NS	NA	NA	No: correlations were -0.13 and NS
Elnaggar et al ¹² n = 56 Lumbar Motion-tracking system	>3 mo 2 wk	Flexion exercise group: 98.5 ± 23.4 Extension exercise group: 93.9 ± 29.8	Flexion exer- cise group: 109.5 ± 22.6 Extension exercise group: 95.2 ± 27.6	Flexion exer- cise group: modified MPQ, 14.1 ± 9.8 Extension exercise group: modified MPQ, 15.9 ± 7.8	Flexion exer- cise group: modified MPQ, 8.9 ± 9.4 Extension exercise group: modified MPQ, 10.6 ± 8.6	Yes (r = -0.24, P = .007)	...	...	...
Mannion et al ³⁹ n = 132 Lumbar Spinal motion analyzer (6-DOF linkage system)	Chronic (>3 mo): fe- male, 10.2 ± 9.9 mo; male, 10.2 ± 9.0 mo 3 mo	NA	NA	...	...	...	...	...	No: changes in mobil- ity were not related to changes in dis- ability (univariate) Changes in mobility did not contribute to or account for changes in disabil- ity (multivariate)
Mellin et al ⁴⁷ n = 456 Lumbar Dual inclinometer	NA 3 wk	NA	NA	...	...	...	NA	NA	No Male: r = 0.00, NS; female: r = 0.23, NS
Johannsen et al ²⁴ n = 27 Multisegment Schober test and sternal elevation	>1 y 5 mo	End group, 5 cm (-3 to 27 was 12.5%); coordina- tion group, 9 cm (-11 to 24 was 12.5%)	NA	Pain score (0- 8) (12.5th percentile): endurance treatment group, 6 (5-8); coordina- tion group, 6 (4-8)	NA	No correlation was found between im- provements in mobility score and im- provements in pain score	Disability (0- 12) (12.5th percentile): endurance treatment group, 6 (3-8); coordina- tion group, 5 (2-8)	NA	No correlation was found between improvement in mobility score and improvement in disability score

Table continues on page A8.

[LITERATURE REVIEW]

APPENDIX D

Study, n, Measurement Type, Measurement Tool	Length of LBP, Time Between Measures	Movement Reported for Cohort Studies ^a		Pain Reported for Cohort Studies ^a			Activity Limitation Reported for Cohort Studies ^a		
		Pre	Post	Pre	Post	Related?	Pre	Post	Related?
Mellin et al ⁴⁵ n = 476 Multisegment Schober test, finger- tip, compass	>2 y 1.5 y	NA	NA	...	...	...	...	...	Male, no; female, yes Male: $r = 0.06$, $P > .05$ Female: $r = 0.39$, $P < .001$
Mellin et al ⁴⁶ n = 194 Multisegment Inclinometer and compass	Chronic 12 mo	NA	NA	...	...	...	...	...	Male, no; female, yes Female: $B = 0.29$, $P < .05$ Male: $P > .1$
Other movement parameters									
Mieritz et al ⁴⁹ n = 199 Lumbar Spinal motion analyzer (6-DOF linkage system)	>6 wk (82% for >1 y) 12 wk	NA	NA	NA	NA	Phase-plot area: no ($r = -0.12$, $P > .01$)	NA	NA	Phase-plot area: no (r $= -0.13$, $P > .01$)
Loisel et al ³⁵ n = 112 Lumbar Spinoscope and EMG	Subacute, from work incidents, absent from work for >4 wk 12, 24, 52 wk	NA	NA	...	...	...	NA	NA	No correlations between changes in ODI and spino- scope scores

Abbreviations: CI, confidence interval; DOF, degree of freedom; EMG, electromyography; FRR, flexion-relaxation ratio; HFQ, Hannover Functional Questionnaire; LBP, low back pain; MPQ, McGill Pain Questionnaire; NA, not available; NRS, numeric rating scale; NS, not significant; NSLBP, nonspecific low back pain; ODI, Oswestry Disability Index; PSC, Patient Specific Complaints; PSFS, Patient-Specific Functional Scale; QBPDS, Quebec Back Pain Disability Scale; RMDQ, Roland-Morris Disability Questionnaire; ROM, range of motion; sEMG, surface electromyography; VAS, visual analog scale.

^aValues are mean or mean $\pm$ SD.